# PRODUCT USER MANUAL

## For
## OMI_CLIMATE_TEMPSAL_BALTIC_Stz_trend

**Issue: 4.0**

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# RECORD TABLE

| Issue | Date | § | Description of Change | Author | Validated By |
|---|---|---|---|---|---|
| 1.0 | 2018/09/06 | all | Creation of the document | U. Raudsepp<br>I. Maljutenko | P. Lagemaa |
| 1.1 | 2019/02/21 | | Revision | U. Raudsepp<br>I. Maljutenko | P. Lagemaa |
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# GLOSSARY AND ABBREVIATIONS

| | |
|---|---|
| **MOI** | Mercator Ocean International |
| **NetCDF** | Network Common Data Form |
| **OMI** | Ocean Monitoring Indicator |
| **PUM** | Product User Manual |
| **QUID** | Quality Information Document |

# DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

# 1) INTRODUCTION

## a) Summary

This document is the user manual for the Copernicus Marine Service Ocean Monitoring Indicator (OMI) OMI_CLIMATE_TEMPSAL_BALTIC_Stz_trend. It contains one dataset:

- omi_climate_tempsal_baltic_stz_trend

The OMI provides a timeseries of Baltic Sea area averaged subsurface salinity trends which contains single profile of subsurface salinity trends at different depth levels ranging down to 250 m. The trend is found based on Baltic Sea reanalysis product BALTICSEA_MULTIYEAR_PHY_003_011.

The product is published on the Copernicus Marine Service dissemination server after automatic and delayed-mode quality controls. The product is available online and disseminated through the Copernicus Marine Service Information System. Files downloaded are in NetCDF format and follow CF-1.7 convention.

Disclaimer: The quality of the product may vary during the proposed time series depending on the possible update of the system.

The OMI is further described in the corresponding Quality Information Document (QUID).

# 2) DESCRIPTION OF THE PRODUCT SPECIFICATION

## a) General Information

| | |
|---|---|
| **Product Lines** | OMI_CLIMATE_TEMPSAL_BALTIC_Stz_trend |
| **Geographical coverage** | Baltic Sea |
| **Variables** | subsurface salinity tendency |
| **Available time series** | From 1993 to Year-1 |
| **Temporal resolution** | Integrated |
| **Horizontal resolution** | Integrated |
| **Integration depth (for example)** | 0 – 250m |
| **Format** | NetCDF4 |

## b) Production System Description

The horizontally averaged subsurface salinity trend has been calculated using the results from Copernicus Marine Service BALMFC Reanalysis product BALTICSEA_MULTIYEAR_PHY_003_011. The linear trend has been calculated from yearly time series of horizontally averaged salinities over the Baltic Sea domain (13 °E - 31 °E; 53 °N - 66 °N; excluding the Skagerrak strait). The upper and lower uncertainty bounds are calculated as the 95% confidence intervals of the yearly trend values.

## c) Dataset details

| |
|---|
| **omi_climate_tempsal_baltic_stz_trend** (subsurface salinity trend) contains the following variables: |
| **so_trend** [365*24*3600*1e-3 s-1]<br>long_name: Horizontally averaged subsurface salinity trend<br>standard_name: tendency_of_sea_water_salinity<br><br>**so_trend_upper95p** [365*24*3600*1e-3 s-1]<br>long_name: Upper 95% confidence bound for the horizontally averaged subsurface salinity trend<br>standard_name: tendency_of_sea_water_salinity<br><br>**so_trend_lower95p** [365*24*3600*1e-3 s-1]<br>long_name: Lowe 95% confidence bound for the horizontally averaged subsurface salinity trend<br>standard_name: tendency_of_sea_water_salinity<br><br>**time** [days since 1950-01-01 00:00:00]<br>long_name: time<br>standard_name: time |

# 3) FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

# 4) FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

Files nomenclature when using the get command in the Copernicus Marine Toolbox or using the Browse File is as follows:

<datasetname>_P<productiondate>.nc

Where :
<datasetname> is the dataset reference
<productiondate> is the production date when the file was generated
.nc is the standard NetCDF filename extension.

# 5) REFERENCES

Quality Information Document (CMEMS-BAL-OMI-QUID-CLIMATE-TEMPSAL-Stz-trend)